

LESSON 107 (0.7 GROUND and 1.3 DUAL)

Lesson Objective

During this lesson, the student will apply the techniques practiced in the training device in the aircraft. The student will be introduced to tracking along a road to increase the students understanding of wind drift and wind correction that can later be correlated to the traffic pattern and approach and landing. The student will demonstrate straight and level flight at various airspeeds and configurations with instructor guidance to enhance his/her understanding of aircraft controls and the effects power and configuration have on the aircraft state. The student will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed.
 - Chapter 7 of the Airplane Flying Handbook - Ground Reference Maneuvers:
 - Drift and Ground Track Control.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 02 - Ground Operations.
 - Chapter 03 - Basic Flight Maneuvers.
 - Chapter 06 - Takeoffs and Departure Climbs.
 - Chapter 07 - Ground Reference Maneuvers.
- Warrior Maneuvers Manual.

Lesson Content (Ground)

Review Fundamentals from Lesson 106

- Quiz student on taxiing procedures and straight and level flight from previous AATD lesson.

Taxiing Procedures

- Quiz the student on the before taxi check.
- Hand stays on the throttle.
- Heels on the floor (move feet up on the pedals when braking).
- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed with an airport diagram (except ramp).

Enhanced Private Pilot Certification Course**Tracking Along a Road**

- Go over chapter 7 (Ground Reference Maneuvers - Drift and Ground Track Control) of the Airplane Flying Handbook.
- Review how to track along a road with wind correction.

Straight and Level Flight Techniques

- Pitch + Power = Performance.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Set pitch THEN power THEN trim.
- Utilize the HSI and inclinometer to maintain bank.
- Utilize the Altimeter and VSI to maintain altitude.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)**Checklist Procedures**

- Re-emphasize the importance of using checklists.
- Have the student run through all of the checklists.
- Assist if needed.
- Have the student go through the “passenger briefing” and the “engine fire during start brief”.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command’s responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.
 - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Taxiing Procedures

- Have the student perform the before taxi check.
- Hand stays on the throttle.
- Heels on the floor (move feet up on the pedals when braking).
- Assist with wind corrections.
- Demonstrate with instrument checks.
- Assist with centerline and speed.
- Assist with runup.
- Ensure that every taxi is briefed with an airport diagram (except ramp).

Review Effects of Controls as Appropriate

- If needed, refer to the lesson plan for 105.

Tracking Along a Road

- Use a straight-line ground reference and have the student track it.
- Ensure the student is correcting for wind drift.

Straight and Level Flight at Cruise Speed

- Demonstrate entering and maintaining straight and level flight.
- Pitch + Power = Performance.
- Set pitch THEN power THEN trim.
- Straight and level sight picture + 2300 RPM = 90KTS straight and level.
- Utilize the HSI and inclinometer to maintain pitch.
- Utilize the Altimeter and VSI to maintain altitude.

Straight and Level Flight at Selected Airspeeds

- For changes greater than 20 knots:
 - Full power or idle power.
 - Anticipate by 5 knots.
- For changes less than 20 knots:
 - 100 RPM change = 5 knots change

Straight and Level Flight with Flaps

Flap Settings / Effects	Lift	Nose	Stabilator(initially)	Drag	Airspeed	Stabilator (due to drag)
0 - 10 Degrees	↑↑↑	↑↑	Forward	↑	↓	Backward
10 - 25 Degrees	↑↑	↑↑	Forward	↑↑	↓↓	Backward
25 - 40 Degrees	↑	↑	Forward	↑↑↑	↓↓↓	Backward
40 - 25 Degrees	↓	↓	Backward	↓↓↓	↑↑↑	Forward
25 - 10 Degrees	↓↓	↓↓	Backward	↓↓	↑↑	Forward
10 - 0 Degrees	↓↓↓	↓↓	Backward	↓	↑	Forward

Enhanced Private Pilot Certification Course**Postflight Procedures**

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 108.
- Assign homework:
 - Read chapter 3 of the Airplane Flying Handbook - Basic Flight Maneuvers.
 - Level turns.
 - Climbs and climbing turns.
 - Descents and descending turns.

Completion Standards

This lesson is complete when the student is able to:

1. Correctly perform previously learned maneuvers with little instructor guidance.
2. Have a weather briefing and a completed TOLD card prepared for the lesson, which will now become standard for all remaining lessons.
3. Demonstrate proper sequential checklist procedures for before and after start, before and after taxi, run-up and takeoff.
4. Demonstrate an understanding of straight and level flight techniques in the aircraft
5. Explain the relationship between power, airspeed, and angle of attack at various speeds.